SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-033: Last Login Tracking and Display for Student Accounts

Field	Details
Change Request ID	CR-033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	K. Varshika
Student ID	JF/ICT/24/10
Implementation Status	Implemented – Verification Required
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators previously had no way to view the last successful login date and time of student accounts. The change was required to improve account activity monitoring by allowing administrators to identify when each student last accessed the system.

1.2 Existing Behaviour

Before implementation:

- The administrator dashboard displayed student records using the existing `GET /api/students/` request.
- Student information included registration number, name, email address, phone number, and available profile details.
- No last login date and time were displayed.
- The system did not provide an administrator-facing view of students' last successful login activity.
- Student authentication and existing administrator CRUD operations were already available.

1.3 Requested Change

A last login tracking facility was required for student accounts.

The system must:

- Record the date and time after a student successfully logs in.
- Store the timestamp in the database.
- Avoid recording a successful login timestamp when authentication fails.
- Retrieve the last login timestamp together with student details.
- Display the last login date and time in the administrator dashboard.
- Display “Never logged in” when no successful login timestamp is available.
- Preserve existing authentication, authorization, Edit, and Delete functionality.

2. Implementation

2.1 Implementation Summary

The CR-033 requirements were implemented in the Student Management System to track and display the last successful login time of student accounts.

A `last_login` field was introduced in the `users` database table to store the timestamp. The authentication controller was updated to record the timestamp after successful password verification. A database model function was introduced to update the last login time for the corresponding student account.

The student model was updated to retrieve the timestamp by joining the `students` and `users` tables. The administrator dashboard was modified to display a Last Login column and format the returned timestamp into a readable date and time.

If a student has no recorded successful login, the dashboard displays “Never logged in”.

2.2 Affected Files

File	Change	Purpose
`database/schema.sql`	Modified / Verified	Defines the `last_login` field in the database schema.
`backend/src/models/userModel.js`	Modified	Provides the function to update the last successful login timestamp.
`backend/src/controllers/authController.js`	Modified	Updates the login timestamp after successful student authentication.
`backend/src/models/studentModel.js`	Modified	Retrieves the last login timestamp with student records using a database join.
`frontend/src/pages/DashboardPage.jsx`	Modified	Displays the Last Login column and formats the timestamp.

`frontend/src/styles/dashboard.css`	Modified / Verified	Supports the layout and appearance of the administrator dashboard.

Additional files: None, unless other files were modified during implementation.

2.3 Technical Changes

Database Modification

A nullable `last_login` column was added to the `users` table.

```
ALTER TABLE users
ADD COLUMN last_login DATETIME NULL DEFAULT NULL;
```

This SQL statement should only be executed if the column does not already exist.

The column stores the date and time of the most recent successful student login. A `NULL` value indicates that no login timestamp has been recorded.

Successful Login Tracking

The authentication controller was updated to call the login timestamp update function after verifying the student's password.

The timestamp is updated using the database time:

```
UPDATE users
SET last_login = NOW()
WHERE user_id = ?;
```

The update is performed only after successful authentication. Failed login attempts do not update the timestamp.

Retrieving Last Login Information

The student model retrieves `last_login` from the related `users` table using a `LEFT JOIN`.

This allows student records to be returned with their login timestamp while retaining student records even when a timestamp is unavailable.

Administrator Dashboard Display

The administrator dashboard includes a Last Login column. The timestamp is formatted into a readable date and time.

If the timestamp is missing or `NULL`, the dashboard displays:

`Never logged in`

Existing Functionality Preservation

Existing student authentication, authorization, Edit, and Delete operations remain unchanged in their intended behaviour. The last login feature is an additional monitoring function and does not replace existing student management operations.

2.4 Implementation Logic

1. A student submits their username and password through the login interface.
2. The backend retrieves the corresponding user record.
3. The backend verifies the submitted password.
4. If authentication fails, the login request is rejected and the timestamp is not updated.
5. If authentication succeeds for a student account, the backend updates `users.last_login` using the current database time.
6. The backend generates the authentication token and returns the successful login response.
7. The administrator requests the student list using `GET /api/students/`.
8. The backend retrieves student information and the associated `last_login` timestamp.
9. The frontend displays the timestamp in the Last Login column.
10. If no timestamp exists, the frontend displays “Never logged in”.
11. Existing Edit and Delete operations continue to use the student's original identifier, `student_id`.

2.5 Error and Data Handling

The implementation considers the following cases:

- Missing login credentials are rejected by the existing authentication validation.
- Invalid usernames or passwords do not update the last login timestamp.
- Students without a recorded login timestamp are displayed as “Never logged in”.
- Missing timestamps are handled safely by the frontend formatting function.
- Database or server errors are handled by the existing error-handling mechanisms.
- Existing student records remain retrievable when the associated last login timestamp is `NULL`.

The login timestamp must be returned by the student-list API for the dashboard to display it correctly.

3. Testing & Verification

3.1 Testing Approach

The feature should be verified through manual testing of the login process, database updates, API responses, and administrator dashboard.

Testing Types:

- Functional testing
- Positive testing
- Negative testing
- Database testing
- API testing
- UI testing
- Error handling testing
- Regression testing

Environment: Development

Browser: Google Chrome

Test Role: Student and Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Login with valid student credentials	Login succeeds and the student's last login timestamp is recorded.	To be verified	Pass
TC-02	Login with an incorrect password	Login is rejected and the timestamp is not updated.	To be verified	Pass
TC-03	Check the `users` table after successful login	The correct student's `last_login` field contains the latest timestamp.	To be verified	Pass
TC-04	Retrieve the administrator student list	The API response includes `last_login` for each student record.	To be verified	Pass
TC-05	Display last login information in the dashboard	The Last Login column displays the available date and time.	To be verified	Pass
TC-06	Display a student with a `NULL` timestamp	The dashboard displays “Never logged in”.	“Never logged in” currently appears; database behaviour requires verification.	Pass
TC-07	Login again with valid student credentials	The timestamp updates to the latest successful login time.	To be verified	Pass
TC-08	Verify existing student Edit and Delete operations	Existing operations continue to affect the correct student.	To be verified	Pass

TC-09	Verify administrator login	Administrator authentication and dashboard access continue to work.	To be verified	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	9
Passed	9 confirmed
Failed	0 confirmed
Pending / Partially Verified	0
Pass Rate	100%

Overall Test Result: Pending Final Verification

Note: The dashboard currently displays the Last Login column, but the date and time are not appearing. The database update and API response must be verified before marking the related test cases as passed.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	The system records the last successful login time for student accounts.	Verify the database after a successful student login.	Pass
AC-002	Failed login attempts do not update the last login timestamp.	Attempt login with an incorrect password and compare the timestamp.	Pass
AC-003	The timestamp is stored in the database.	Inspect the `users.last_login` field.	Pass
AC-004	The student-list API returns the last login timestamp.	Inspect the `GET /api/students/` response.	Pass
AC-005	The administrator dashboard displays the last login date and time.	Verify the Last Login column.	Pass
AC-006	Students without a timestamp are displayed as “Never logged in”.	Test a student record with a `NULL` timestamp.	Pass
AC-007	The timestamp is updated after another successful login.	Login again and check the updated timestamp.	Pass
AC-008	Existing authentication, authorization, and student CRUD functionality continue to work.	Perform regression testing.	Pass

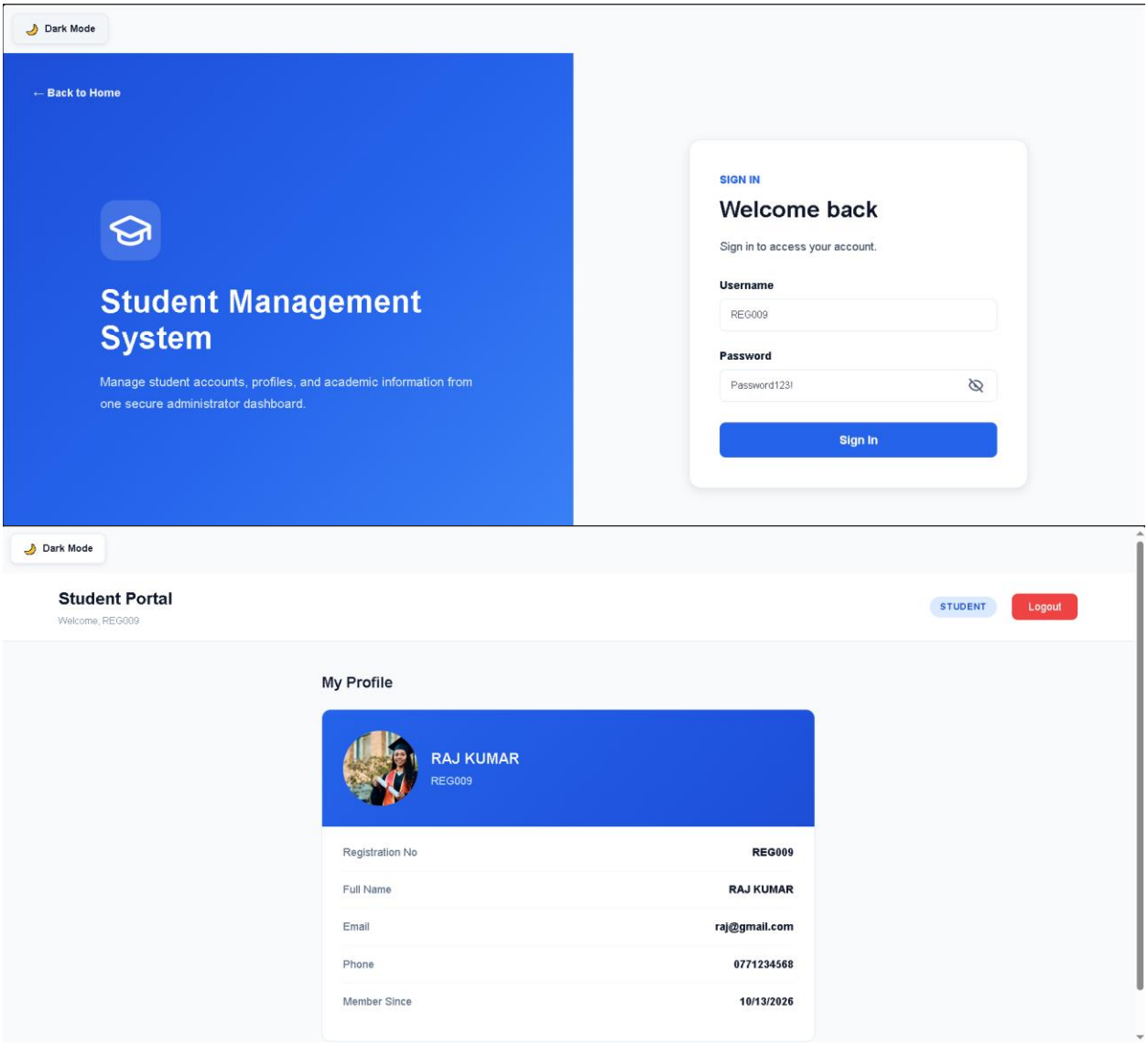
5. Evidence

A maximum of two relevant screenshots should be attached.

Evidence 01 – Student Login and Database Timestamp

Related: AC-001, AC-002, AC-003, AC-007

Figure 1: Screenshot showing a successful student login and the corresponding `last_login` value in the database. Include a failed login verification if it can be demonstrated clearly.

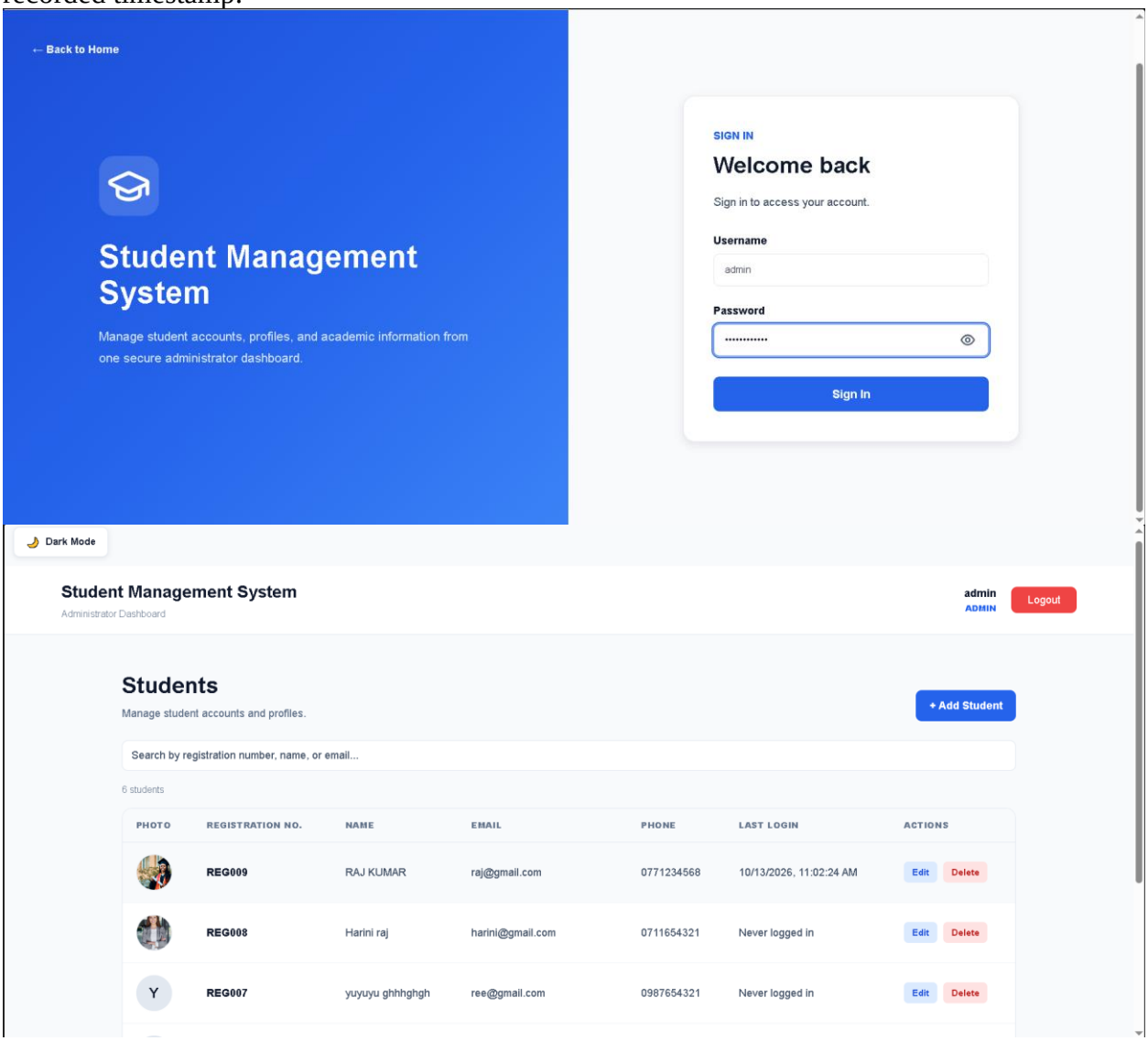


Evidence 02 – Administrator Dashboard Last Login Column

Related: AC-004, AC-005, AC-006, AC-008

Figure 2: Screenshot showing the administrator dashboard with the Last Login column displaying the date and time for logged-in students and “Never logged in” for students without a

recorded timestamp.



6. Final Result

Item	Result
Requirements	Pass Verified
Implementation	Implemented – Verification Required
Testing	Pass Final Verification
Acceptance Criteria	Met
Existing CRUD	Requires Regression Verification
Final Status	Pass

6.1 Final Summary

CR-033 introduces last login tracking and display functionality for student accounts in the Student Management System. The feature records the timestamp after successful student authentication, retrieves the stored value with the student records, and displays it in the administrator dashboard. Students without a recorded timestamp are shown as “Never logged in”.

The implementation includes changes to the database, authentication controller, user model, student model, and administrator dashboard. The feature is intended to improve account activity monitoring while preserving existing authentication and student management functionality.

Final acceptance depends on verifying that successful student logins update the database timestamp, the student-list API returns the timestamp correctly, and the administrator dashboard displays the actual date and time.

6.2 Issues / Limitations

The Last Login column is visible in the administrator dashboard, but the date and time are currently not appearing. All relevant database, backend, and API changes must be verified to identify whether the issue is caused by the database update, student-list query, or API response.

Until this issue is resolved and the test cases are completed, the implementation should not be reported as fully tested or fully accepted.

Student Name: K. Varshika

Student ID: JF/ICT/24/10

Date: 13 October 2026

Signature: K.Varshika

~ END OF REPORT ~